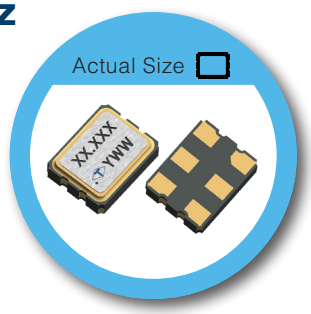
VA-M Type High Frequency up to 1.5GHz
3.2 x 2.5 mm SMD Differential Output Crystal Oscillator



RoHS Compliant

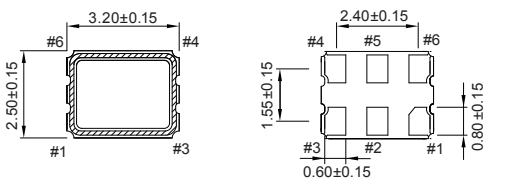
FEATURE

- Low power supply voltage: 3.3, 2.5 supply options
- Differential output : LVPECL, LVDS
- Frequency support from 10MHz to 1.5GHz
- Low phase jitter typical: 0.6 ps RMS from 12kHz to 20MHz
- Wide frequency control range
- Pb-free/RoHS compliant
- Tri-state enable/disable function
- Temperature range: -40 to 85 °C operation

TYPICAL APPLICATION

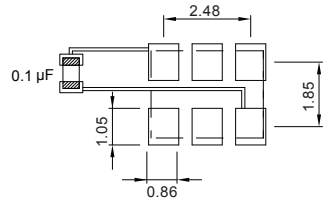
- High-Speed Gigabite Ethernet, Fiber Channel, Storages Area Network, SONET
- Enterprise Server, SAS/SATA
- Microprocessors/DSP/FPGA
- Broadband Access
- Smart Grid

DIMENSION (mm)



Pin#	Function
1	Vcon
2	Tri-State
3	GND
4	Output
5	Comp. Output
6	Supply Voltage

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter			LVPECL 3.3 V Min.	LVPECL 3.3 V Max.	LVPECL 2.5 V Min.	LVPECL 2.5 V Max.	LVDS 3.3 V Min.	LVDS 3.3 V Max.	LVDS 2.5 V Min.	LVDS 2.5 V Max.	unit
Supply Voltage Variation (VDD)			VDD-10%	VDD+10%	VDD-5%	VDD+5%	VDD-10%	VDD+10%	VDD-5%	VDD+5%	V
Frequency Range			10	1500	10	1500	10	1500	10	1500	MHz
Supply Current			–	50	–	45	–	45	–	35	mA
Output Level	Output High		2.27	2.7	1.47	1.9	–	1.6	–	1.6	V
	Output Low		1.45	1.7	0.65	0.9	0.9	–	0.9	–	
Transition Time (10%-90%)	Rise Time / Fall Time		–	1.0	–	1.0	–	1.0	–	1.0	nSec
Duty Cycle			45	55	45	55	45	55	45	55	%
Startup Time			–	10	–	10	–	10	–	10	mSec
Tri-State mode (Input to Pin 2)		Enable	0.7 x VDD	–	0.7 x VDD	–	0.7 x VDD	–	0.7 x VDD	–	V
		Disable	–	0.3 x VDD	–	0.3 x VDD	–	0.3 x VDD	–	0.3 x VDD	
Stand by Current			–	18	–	18	–	18	–	18	mA
Output Loading			50Ω into VDD-2V				100Ω				
Phase Noise			*Typ.*	*Max.*	*Typ.*	*Max.*	*Typ.*	*Max.*	*Typ.*	*Max.*	
At VDD=3.3V, fout=250MHz		1kHz offset	-107	–	-107	–	-107	–	-107	–	dBc/Hz
		10kHz offset	-111	–	-111	–	-111	–	-111	–	
		100kHz offset	-114	–	-114	–	-114	–	-114	–	
		1MHz offset	-125	–	-125	–	-125	–	-125	–	
		20MHz offset	-147	–	-147	–	-147	–	-147	–	
RMS Phase Jitter (12kHz to 20MHz)(Fractional mode)			0.8	1.5	0.8	1.5	0.8	1.5	0.8	1.5	pSec
RMS Phase Jitter (12kHz to 20MHz)(Integer mode)			0.6	1.2	0.6	1.2	0.6	1.2	0.6	1.2	pSec

Parameter	Control Voltage Function on Pin 1 3.3 V Min.	3.3 V Max.	2.5 V Min.	2.5 V Max.	unit
Control Voltage Center	1.65		1.25		V
Control Voltage Range	0.3	3	0.25	2.25	
Frequency Pulling Range	±50	±150	±50	±150	ppm
Linearity	5	10	5	10	%
Modulation Bandwidth	10	–	10	–	kHz
VC Input Impedance	1	–	1	–	MΩ

FREQ. STABILITY vs. TEMP. RANGE

Temp. (°C) \ ppm	±25	±50
-10 ~ +60	○	○
-20 ~ +70	○	○
-40 ~ +85	△	○
-40 ~ +105	X	△

* ○: Available △:Conditional X: Not available
* Inclusive of calibration @ 25 °C, operating temperature range, input voltage variation, load variation, aging (1st year), shock, and vibration

Note: not all combination of options are available. Other specifications may be available upon request.